

Thomas Kennedy

linkedin.com/in/tken

github.com/tksoftw

Charlottesville, VA

U.S. Citizen

EDUCATION

University of Virginia

Aug. 2024 - Expected May 2028

B.A. Computer Science; B.A. Mathematics

Charlottesville, VA

GPA: **3.86** / 4.0 CS GPA: **3.94** / 4.0

Relevant Coursework: *Data Structures & Algo, Computer Architecture & OS, Software Development, Adv. Programming Languages, Machine Learning, Probability, Linear Algebra, Stochastic Processes.*

EXPERIENCE

Research Assistant

Oct. 2025 - Present

UVA Biocomplexity Institute

Charlottesville, VA

- Built a **differentiable agent-based simulator** of hospital patient flow in **PyTorch**. Learned parameters via gradient descent on real hospital data. Integrated into a larger **Recurrent neural network (RNN)**.
- Predicted daily hospital patient occupancy with high accuracy (**MAE** $\approx$ **1 patient** over 208 days), forming the core methodology for an upcoming paper.

Teaching Assistant (CS 3130 & CS 2130)

Jan. 2025 - Present

University of Virginia

Charlottesville, VA

- CS 2130: **Head TA**. Manage a staff of **24+ TAs** and organize office hours for **400+ students**. Deliver lectures and direct TAs in lab sections on x86-64 assembly, C, and computer architecture.
- CS 3130: **TA**. Instruct labs and hold office hours for UVA's advanced computer architecture course.

Software Engineer Intern

Jul. - Aug. 2025

Verizon

Basking Ridge, NJ

- Created a **full-stack web application** to automatically generate YAML configuration files for the company's internal Red Hat OpenShift platform, replacing manual entry.
- Reduced engineering workload by up to **6 hours/week** and eliminated **2+ weeks** of vendor dependency wait time per release.

Research Intern

Jun. - Jul. 2025

TRB Capital Management LLC

Charlottesville, VA

- Engineered a robust **C++** interface on top of the **QuickFIX engine** to facilitate standardized, high-speed communication within an automated trading system.
- Created high-level system design visualizations using **Python** and **NetworkX** to simplify complex infrastructure and map out strategic design planning.

Software Engineer Intern

Jul. 2024 - Aug. 2024

Meritarc

Remote

- Directed the software architecture and end-to-end implementation of AI agents for car payment workflows using **Microsoft Semantic Kernel**.

LEADERSHIP

Founder

Mar. 2025 - Present

+EV Club

Charlottesville, VA

- Started a club dedicated to researching and automating online arbitrage opportunities.
- Generated **\$30,000+** in total club profits.

PROJECTS

Lenovo IdeaPad 720s Hackintosh

Feb. 2025 - Present

- Transformed a Lenovo IdeaPad 720s laptop into a fully functional macOS Tahoe workstation.
- Built the EFI boot sequence entirely by hand from the ground up to guarantee system stability.
- Saved over \$300 in hardware costs.

Skills & Interests

Skills: Python, C/C++, Reverse-engineering, Linux, Selenium, x86-64 Assembly, Java, Minecraft PvP

Interests: Data visualization, advantage play, rhythm games, Nintendo games